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AN EXPERIMENTAL INVESTIGATION INTO ASPECTS OF EROSION IN ROCKET MOTOR TAIL NOZZLES

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SUMMARY

Results are given of experiments carried out on a test rig which has been developed to produce damage of a type similar to that found in the nozzles of rocket motors. The experiments were, in the main, concerned with the erosive action of abrasive particles carried by the gas stream, although ablation effects at the nozzle wall due to wall heating were also investigated. The results indicated that the degree and pattern of damage within a nozzle of a given material depended on (a) the particle velocity, (b) the particle flow pattern, (c) the particle distribution in the flow area and (d) the size and type of the particles. The angle of attack of the particle stream on the wall is an important factor in erosion. For hard brittle substances maximum damage occurs with normal impact, whereas with ductile materials maximum erosion takes place when the angle of attack is small. This angle for maximum erosion appears to depend only on the physical properties of the material under attack. Experimental data are given showing that high local temperatures are induced in the surface layers of the material under attack. For a given size of particle it was found that the particle velocity in the nozzle efflux remained virtually constant as the ratio of particle mass flow rate to air mass flow rate was increased up to a certain critical value. Beyond this value particle velocity and erosion damage decreased. During the tests it was observed that for all particles used the particle velocity in the nozzle efflux was very much less than the theoretical efflux velocity of the unladen air stream.

INTRODUCTION

Damage in rocket motor tail nozzles may occur by thermal shock, oxidation and/or vaporisation of a component of the nozzle wall material, ablation of surface material weakened by the high operating temperature or by removal of wall material due to the action of entrained solid or liquid particles. A study was made of the nozzle wall damage obtained during actual firings. Here it was found that, while some tail nozzles had failed due to thermal shock, characterized by a large axial crack in the nozzle wall, most of the damaged nozzles appeared either to have suffered an even erosion around the throat section or deep axial grooves had been gouged out of the nozzle wall at one angular position. It was felt that the abrasive action of solid particles was the cause of the damage, especially in motors containing a high percentage of aluminium in the fuel.

The work described here was of an exploratory nature and was centred mainly on the action of gas-borne particles though some additional work at elevated temperatures on ablation effects was also attempted. The first step was to establish whether, in fact, the type of damage suffered by rocket nozzles could be simulated in simple laboratory tests. Preliminary experiments proved that quite a remarkable rate of erosion could be achieved in the laboratory at atmospheric temperatures and with relatively moderate abrasive flow conditions.

Inital experiments showed that the pattern of damage in the nozzle depended on the distribution of particles over the cross-section at entry to the nozzle. The apparatus was modified so that a comparison could be made of the erosive action for uniform and non-uniform distributions of the abrasive for various nozzle wall materials and for varying velocities of the particle-laden stream. The effect of a blast pipe on these variables was also considered. It was shown that, where the particles within the nozzle had a small velocity component normal to the wall, considerable damage could ensue. This led to a series of experiments in which a hardened steel nozzle was used to accelerate the particles and project them on to flat plate specimens mounted at various angles to the incident stream. In these experiments the influence of the material properties of the specimen on the erosion–angle of incidence characteristics were studied.

To investigate the effect of elevated wall temperature, a 27 kW air heater was added to the test rig. This enabled a study to be made of the influence of wall temperature on ablation in the absence of particles. Subsequently the dependence of erosion on initial wall temperature was investigated.

During experiments on the effect of particle size it was found that a hardened steel nozzle, which had effectively withstood the abrasive action of considerable quantities of large sized particles, was severely damaged by the action of 297 μ aluminium oxide powder. This initiated a series of experiments to determine if the erosion potential of these small particles was entirely due to their greater acceleration through the nozzle. A photographic technique was used to determine the particle velocity at nozzle exit.

APPARATUS

The general arrangement of the test rig is shown in Figs. 1 and 2. Compressed air is passed from a large storage vessel through a 27 kW heater to a nozzle box. The

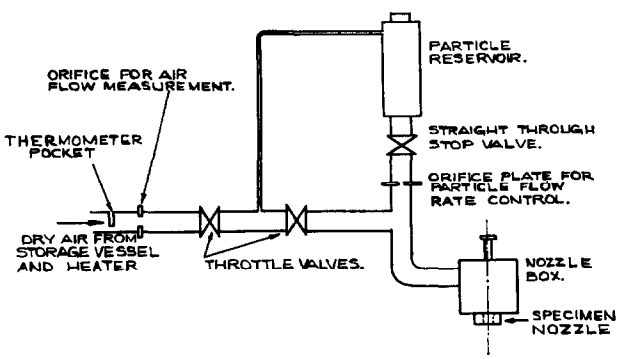


Fig. 1. The experimental rig.

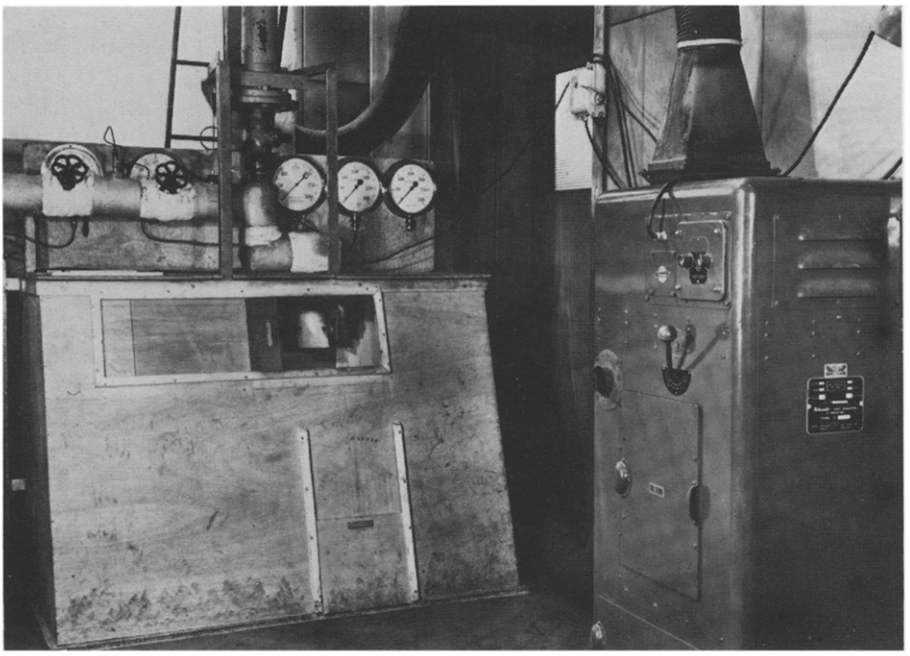


Fig. 2. General view of the experimental rig showing the settling chamber and dust extraction unit.

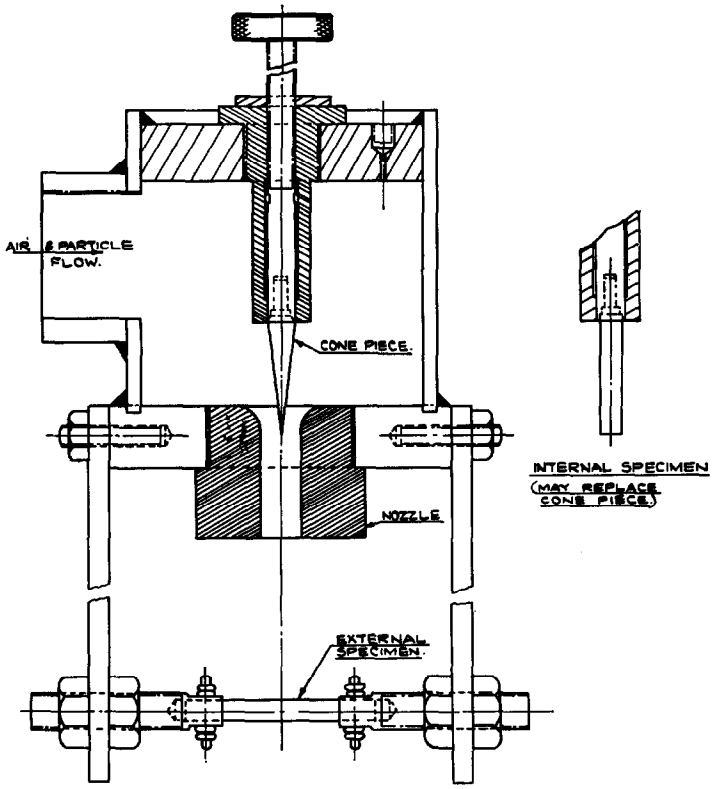


Fig. 3. Nozzle box showing specimen nozzle, cone piece and specimen internal circular insert.

air flow and pressure in the nozzle box are controlled by needle valves in the pipe line. Abrasive particles are contained under pressure in the abrasive reservoir and are allowed to enter the air stream through a control orifice plate, the size of which determines the mass flow rate of the particles. The particle-laden stream is accelerated through the nozzle into a totally enclosed settling chamber. The settling chamber is connected to a dust extraction unit which maintains a slight vacuum in the chamber and draws the particle and air efflux from the nozzle. Details of the nozzle box are shown in Fig. 3. The nozzle is convergent parallel of circular cross-section. The cone piece may be advanced into the nozzle giving a nozzle with a variable exit to throat area ratio. Alternatively, the cone piece may be replaced by a parallel internal piece of circular cross-section running the full length of the nozzle and so forming a nozzle of annular cross-section.

The cone piece, parallel internal piece and nozzle are subjected to the action of the particle-laden stream and may thus be used as specimens in the investigation. Alternatively, the nozzle arrangement may be used merely as an accelerator for the abrasive stream. In this case the stream impinges on external specimens placed in the nozzle efflux some distance from the nozzle exit. In early experiments these external specimens were circular rods, as shown in Fig. 3. Later the rods were replaced by flat plate external specimens which could be mounted at various angles to the incident stream. The arrangement is shown in Fig. 4.

The following types and sizes of abrasive particles have been used.
Angular chilled iron grits of 500 and 250 μ mean particle size.
Angular aluminium oxide grits of 710, 500, 297 and 210 μ mean particle size.
Silicon carbide grit of 500 μ mean particle size.



Fig. 4. Test arrangement for external flat plate specimens.

EARLY EXPERIMENTAL WORK

Specimen nozzles, internal circular inserts and external circular rods in hardened steel, mild steel, brass, aluminium, tufnol*, carbon and perspex were examined. Results of typical tests are given below.

Tests with external circular specimens

Table I gives the observed weight loss of external rod specimens, each of which

TABLE I

Material	*Weight loss (g)*	*Loss in weight (%)*	*Time taken*
Hard steel	1.2	2.56	10 min
Mild steel	7.14	15.15	10 min
Brass	16.5	32.2	10 min
Aluminium	4.25	26.4	10 min
Tufnol	3.77	46.4	0.5 min
Carbon	3.52	48.8	a few seconds
Perspex	3.22	45.4	about 2 min

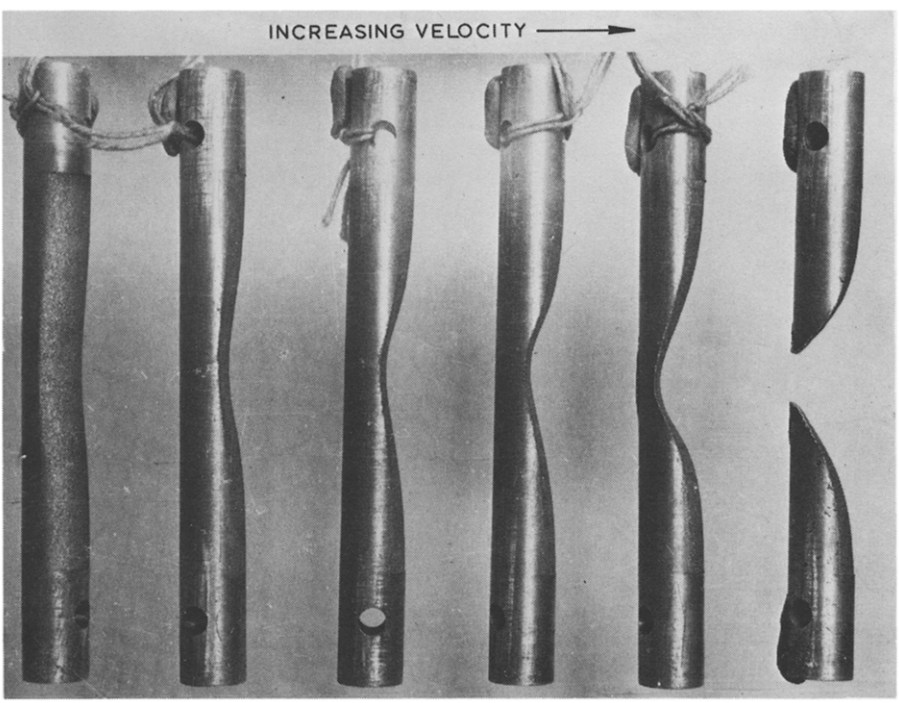


Fig. 5. Progressive erosion of external specimens with increasing air speed.

* Tufnol is a trade name for a thermosetting laminated plastic material produced from layers of paper, cotton, asbestos, or other fillers bonded with a synthetic resin. The type examined here had a paper base. It is widely used as an engineering material, for example, in bearings and in gear wheels.

was subjected to the action of 50 lb. of 500 μ iron grit with a nozzle box pressure of 25 p.s.i.g. These results showed that a measurable degree of erosion could be obtained in a very short time.

Using a hard steel nozzle and cone piece 20 lb. of iron grit was directed at varying velocities on to external aluminium rod specimens. The resulting damage to the rods is shown in Fig. 5. The damage is seen to increase with increasing particle velocity leaving the nozzle.

Tests with internal co-axial specimens

Parallel inserts in various materials were placed inside a hard steel nozzle and each subjected to the action of 20 lb. of 500 μ iron grit. Typical results are shown in Fig. 6. It was found that maximum damage to the internal specimen occurred in a short length downstream of the throat section of the nozzle. The damage was not evenly distributed around the throat circumference but appeared, in all the specimens, as a short axial groove in the vicinity of the throat.

The tests were repeated using 250 μ iron particles and the results are shown in Fig. 7. The pattern of damage was an exaggerated version of that obtained with the larger grit. The short axial hollow near the throat has developed into a deep axial groove extending from the throat to near the end of the specimen. There is also considerable erosion of the insert in the throat region opposite the groove.

These tests indicated (1) that there was an uneven distribution of the particles round the nozzle annulus and (2) that the particles had a small component of velocity normal to the internal specimen at the throat section. An apparatus was designed to

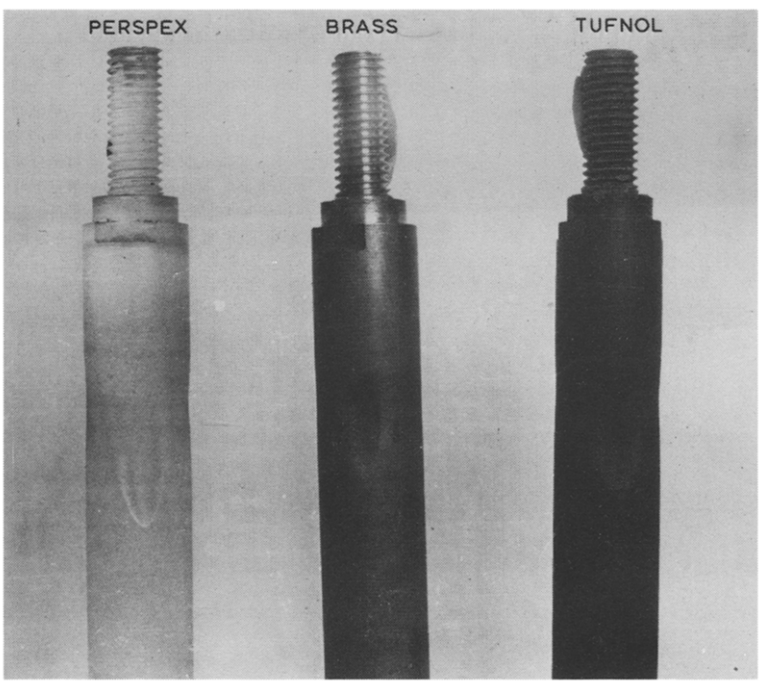


Fig. 6. Moderate grooving of internal specimens of perspex, brass and tufnol by 500 μ iron particles.

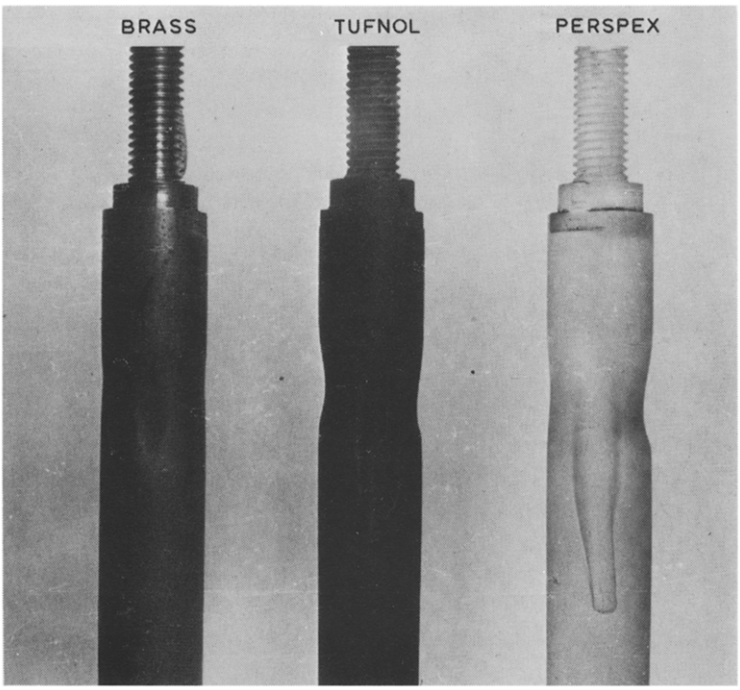


Fig. 7. Extensive grooving of internal specimens of brass, tufnol and perspex by 250 μ iron particles.

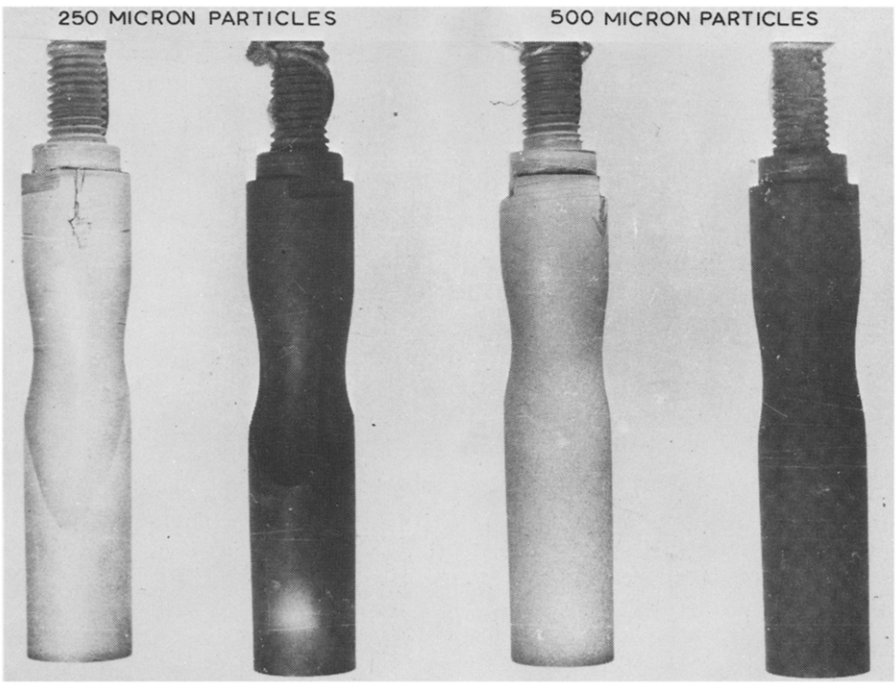


Fig. 8. Uniform erosion of internal specimens, two of tufnol and two of perspex.

measure the grit flow distribution around the annulus, which confirmed that the nozzle segments nearest the air supply pipe in the nozzle box were passing a much higher percentage of the total flow than segments diametrically opposite the air pipe.

Temporary measures were taken to improve the grit distribution and the above tests were repeated with a more even distribution in the nozzle annulus. The results are shown in Fig. 8. The erosion is more even round the specimen and the deep axial grooves obtained previously are absent. Again the smaller particles were found to cause the greatest damage.

Tests were made on nozzles and internal parallel inserts using 500 μ aluminium oxide particles. To study the flow pattern of the particles in the nozzle, various combinations of soft and hard materials for the nozzles and inserts were examined. The

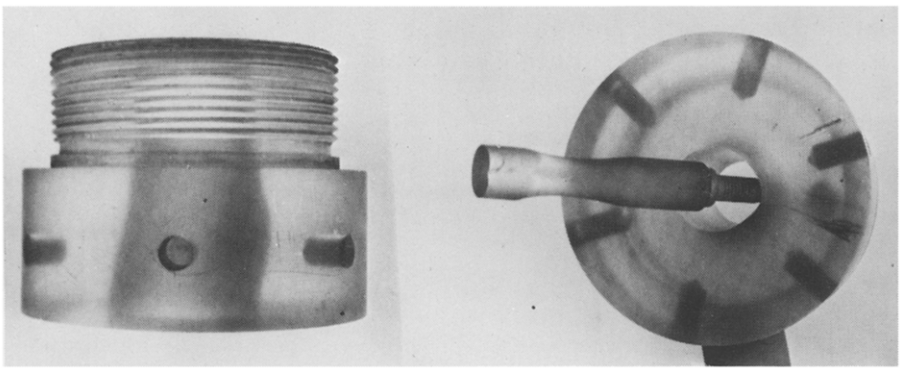


Fig. 9. Perspex specimens–500 μ aluminium oxide particles.

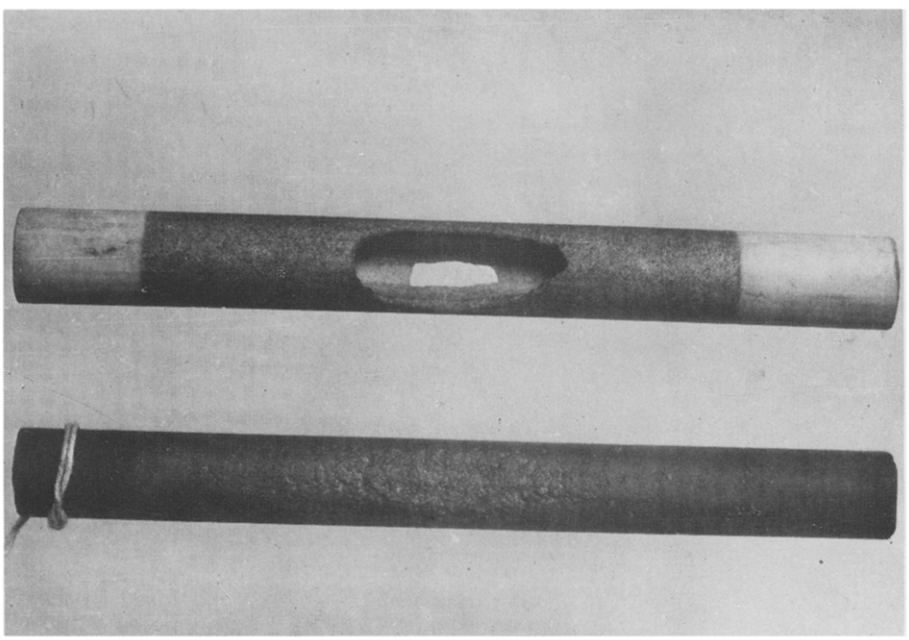


Fig. 10. Erosion of aluminium oxide tubes by alumina powder.

results can be summarised from the test on a perspex nozzle and insert shown in Fig. 9. Here there is a deep axial groove in the nozzle starting at the throat with a larger hollow on the opposite side but nearer the outlet section. These areas of maximum damage to the nozzle profile correspond with the position of deep grooves in the internal specimen.

Further tests with external specimens

The erosion-resistance properties of some extremely hard refractory materials were examined. The specimens were tubes of aluminium oxide, mullite (an aluminium silicate compound) and aluminium porcelain which were tested as external specimens using 20 lb. of 500 μ aluminium oxide particles as the abrasive. Typical results are shown in Fig. 10. In some instances the tubes were pierced on both sides by the jet and the erosion was so severe that the tube finally shattered. In similar tests with hard steel external specimens little erosion damage occurred.

Blast tube effects

To study the effects of the blast tube in the actual rocket motor the arrangement shown in Fig. 11 was used. The hardened steel first stage nozzle acts as a blast

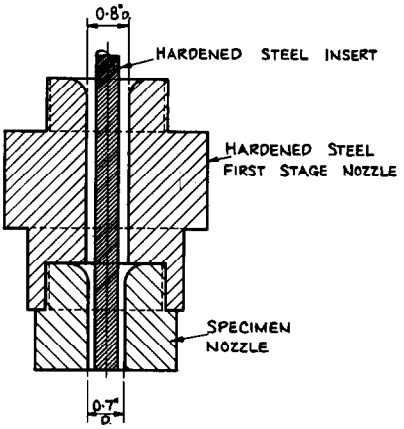


Fig. 11. Two-stage nozzle arrangement giving blast-tube effect.

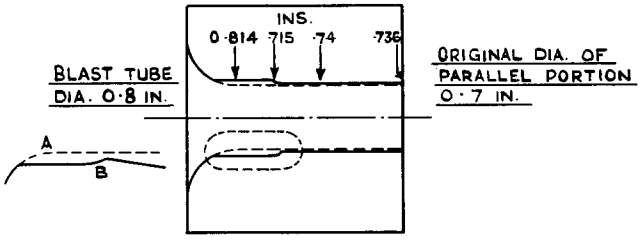


Fig. 12. Profiles of specimen nozzle before and after testing in blast-tube arrangement.

tube. The arrangement gave an area ratio of blast tube to nozzle throat of 1.55 and a blast tube Mach number of 0.41. Damage to a specimen perspex nozzle, which represents the actual nozzle of the rocket motor, is shown in Fig. 12. The results of a number of similar tests showed that

(a) although the distribution of particles entering the blast tube was uneven the erosion around the specimen nozzle was uniform. This indicates that the blast pipe serves to re-distribute the particles more evenly.

(b) the original throat of the specimen nozzle has been eroded away and a new throat formed further down the parallel portion. At the original throat the diameter of the specimen nozzle after erosion is approximately equal to the diameter of the blast tube. This means that the high velocity particles leaving the blast tube do not follow the air streamlines but impinge directly on to the choke section of the specimen nozzle.

Tests with smaller aluminium oxide particles

In most of the previous tests on external specimens a hard steel nozzle and insert were used to accelerate the abrasive stream. In the tests considerable quantities of large size (above 500 μ) aluminium oxide and iron particles had passed through the nozzle. Further tests were carried out to study the effect of particle size by using 297 and 210 μ aluminium oxide particles. It was found that after a few tests with these small particles the hard steel nozzle and insert were extensively damaged. Typical damage to such a hard steel nozzle and insert is shown in Fig. 13.

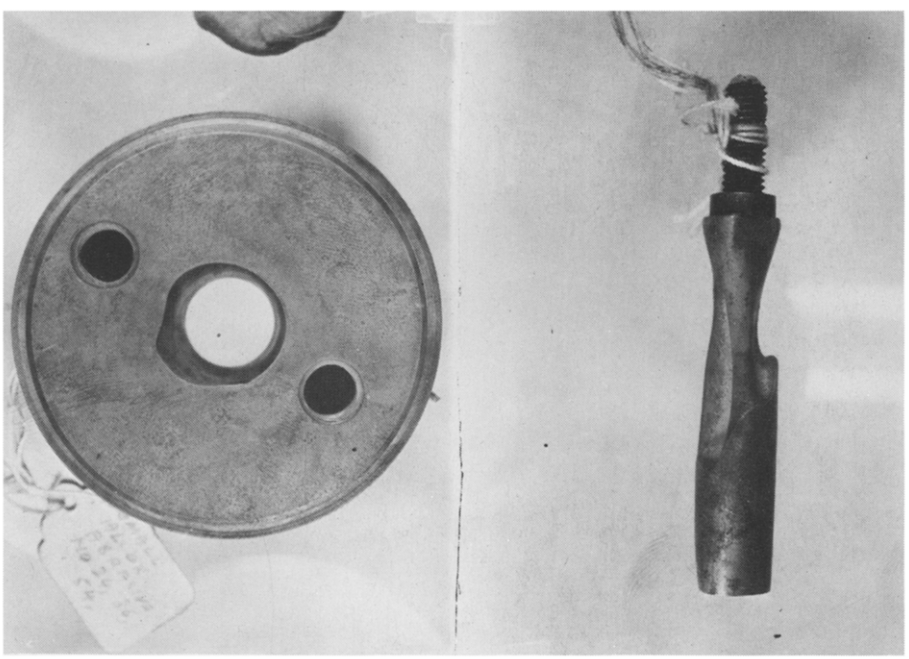


Fig. 13. Damage to a hard steel nozzle and insert by 297 μ aluminium oxide particles.

FURTHER TESTS

Following these preliminary tests, the apparatus was modified and the following series of tests were carried out in sequence.

The determination of the effect of angle of attack.

The observation of the wall surface temperature during bombardment by particles.

The effect on erosion damage of the initial temperature of the surface under attack.

The determination of the particle efflux velocity.

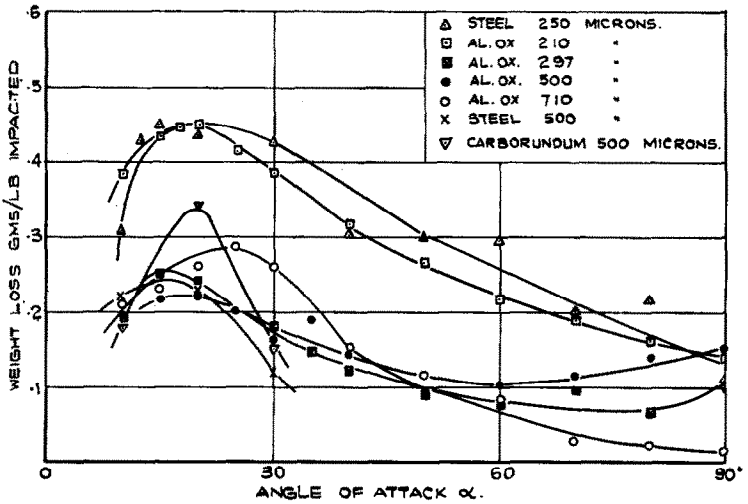


Fig. 14. Variation of erosion for aluminium plates.

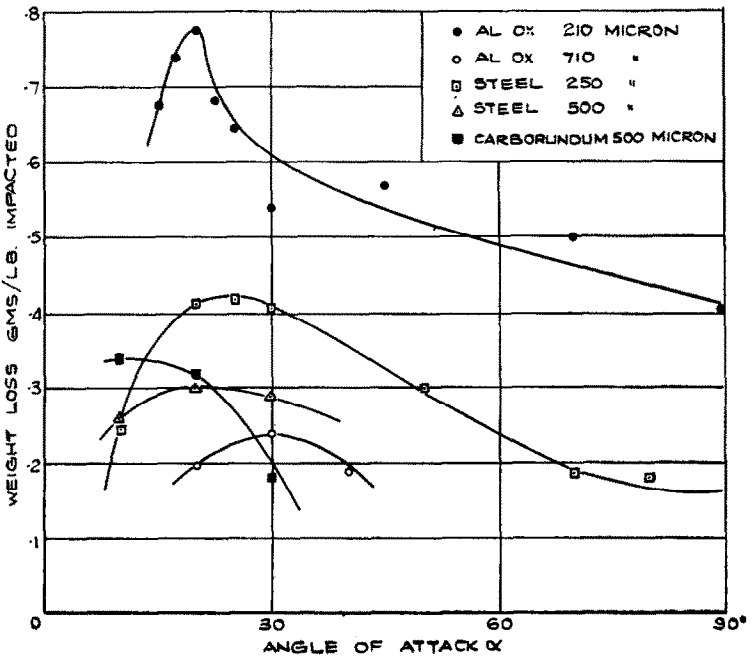


Fig. 15. Variation of erosion for mild steel plates.

The assessment of the resistance offered to a particle attack by the "anti-erosion" coatings in current use in rocket motors.

The results obtained were as follows.

The determination of the effect of angle of attack

In this study, flat external specimens were mounted on a holder, as shown in Fig. 4. Tests were made using different sizes and types of grit on plates of different materials mounted at various angles to the incident stream. The results of tests on aluminium external plates are shown in Fig. 14. They show that the angle for maximum

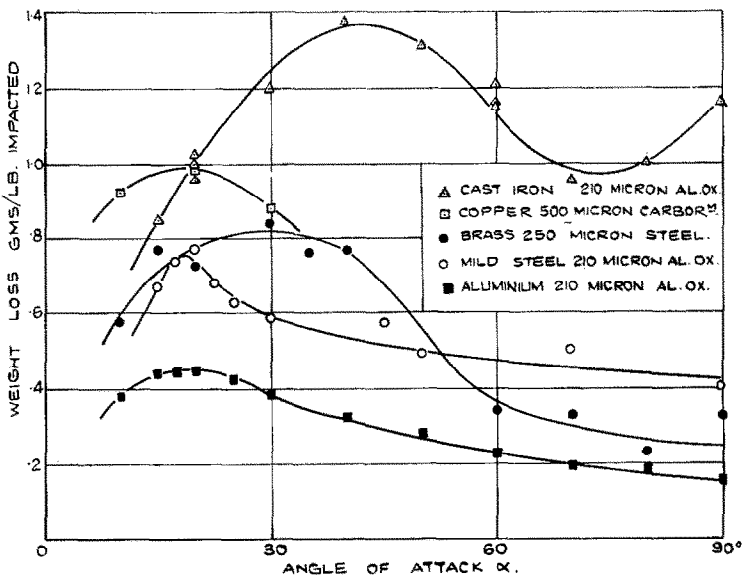


Fig. 16. Variation of erosion for different plate materials.

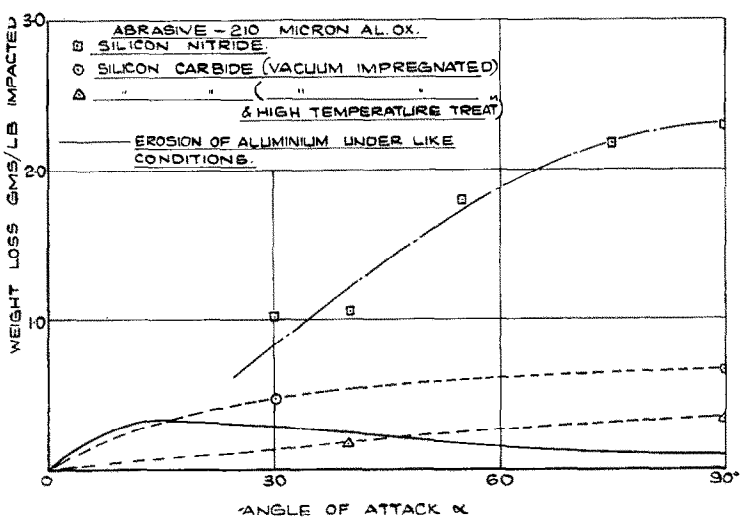


Fig. 17. Variation of erosion for hard ceramic materials.

erosion is about 20° and that this angle appears to be independent of the type of abrasive particle or its velocity. Figure 15 gives the corresponding results for mild steel plates. In Fig. 16 a comparison is made of the way in which the angle for maximum erosion varies with the specimen material under attack. For the comparatively brittle cast iron plates the angle for maximum erosion is 40° compared with the 20° for the ductile aluminium plates. Typical curves for hard ceramic materials are given in Fig. 17. Here maximum erosion damage occurs at normal attack.

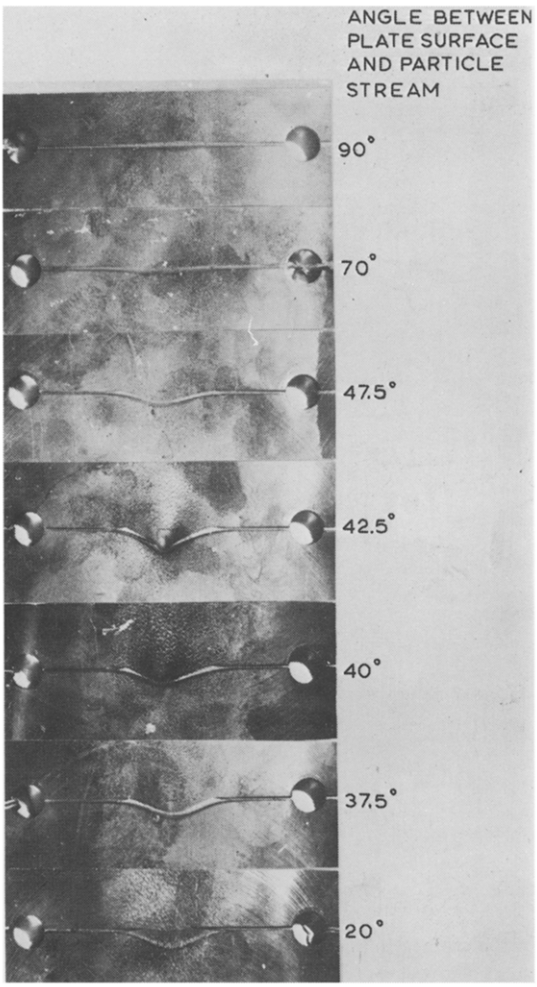


Fig. 18. Series of tests on cast iron plates at different angles of incidence.

Before the brass and cast iron plates were tested, lines about 1/32 in. deep were scribed across the middle of the plates. It was found that for both the ductile brass and the brittle cast iron the lines apparently moved in the direction of the particle flow. The displacement of the line was a maximum at the angle for maximum erosion damage in each case. The results for cast iron specimens are shown in Fig. 18.

To show that the form of erosion within the nozzle is similar to that observed in flat external plate specimens, nozzles and inserts in aluminium, brass and copper were scribed with a number of circumferential grooves and subjected to a particle attack. A typical result is shown in Fig. 19 where the apparent movement of a groove can be seen. By comparison with the external plates this result indicates that the nozzles and inserts are subjected to a particle attack at a small angle.

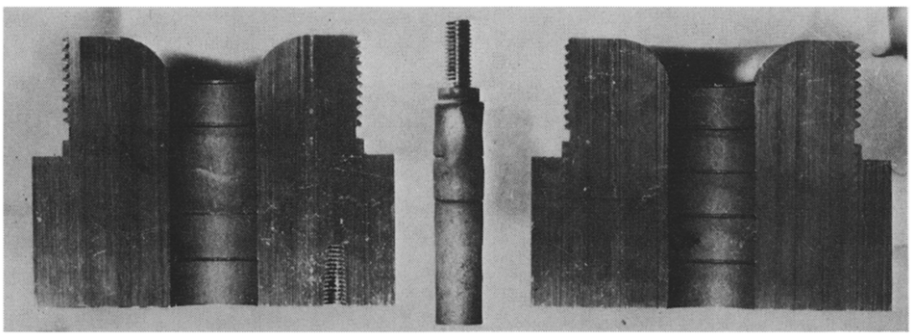


Fig. 19. Brass nozzle and insert.

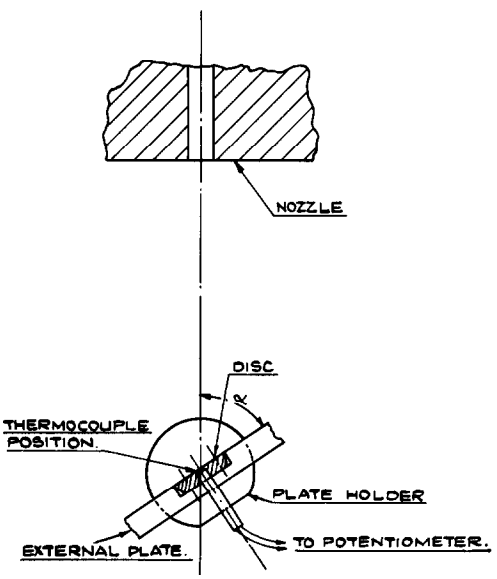


Fig. 20 Arrangement used for observation of the surface temperature during particle attack.

The variation in surface temperature at the wall during erosion

It was observed that the plate temperature rose during particle attack and this prompted a closer investigation of the temperature at the specimen surface during attack. The arrangement shown in Fig. 20 was used. Hardened steel discs were mounted in a modified external plate and a thermocouple was located at 1/64 in. from the exposed surface of the disc. The disc was designed to limit radial heat flow from the disc to the rest of the plate. The discs were subjected to attack by 500 μ and 297 μ

particles. It was found that the rise in temperature during attack was, on average, some 300° F above the initial steady-state temperature. With the larger particles there was a continuous fluctuation in the temperature whereas with the small particles there was a uniform rise in temperature. The fluctuation in temperature with the large particles was attributed to a random bombardment around the thermocouple area and the uniform rise was no doubt due to the more intense particle flux associated with the smaller particles. The duration of each test was determined by the time taken to erode through the 1/64 in. thick layer above the thermocouple. This took place in some 30 sec and it was felt that the actual temperature at the time of impact was much greater than that observed at the thermocouple.

The effect of initial surface temperature on erosion damage

The object here was to simulate more closely the actual conditions during firing. Nozzles and external plate specimens were made in Cerrobend alloy which melts at 158°F. In experiments with air only passing through the apparatus, the temperature of the air was raised in steps and the nozzle examined closely at each increment for signs of initial damage to the surface. Typical results for the nozzles are shown in Figs. 21–23. In each case the nozzle box, at which melting commenced, was around 200°F, and the temperature recorded at a thermocouple placed midway through the parallel portion at 1/16 in. from the wall surface was about 15 degrees below the melting point temperature.

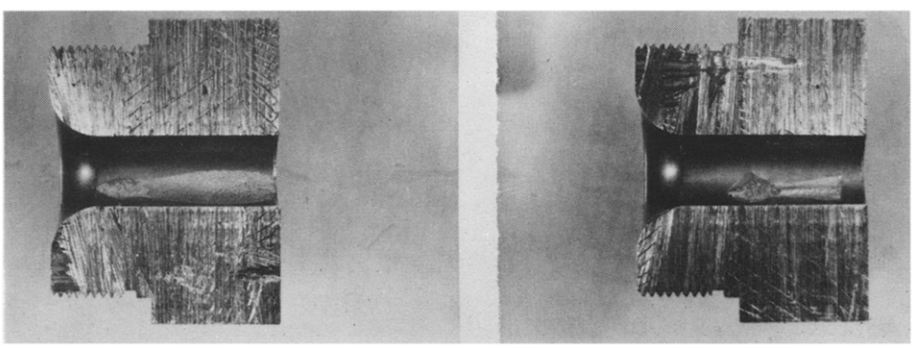


Fig. 21. Cerrobend alloy melting at pressure tapping holes.

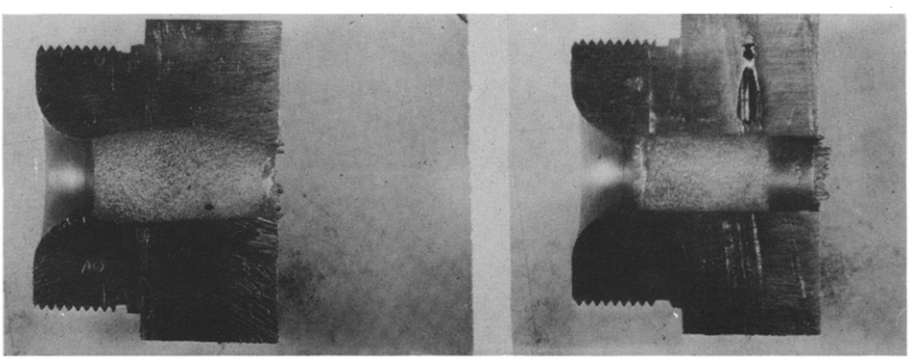


Fig. 22. Cerrobend nozzles melting from the throat.

In Fig. 21 initial melting has occurred at the small pressure tapping holes. The material removed from around the holes has resolidified on the cooler parts of the nozzle. In Fig. 22, where there were no tapping holes, it will be noted that initial melting takes place from the throat section and that the choke is not damaged. For the nozzle in Fig. 23 a fissure was observed in the choke surface prior to the test. Initial melting developed from this irregularity at a nozzle box temperature below that at which melting from the throat section would normally ensue.

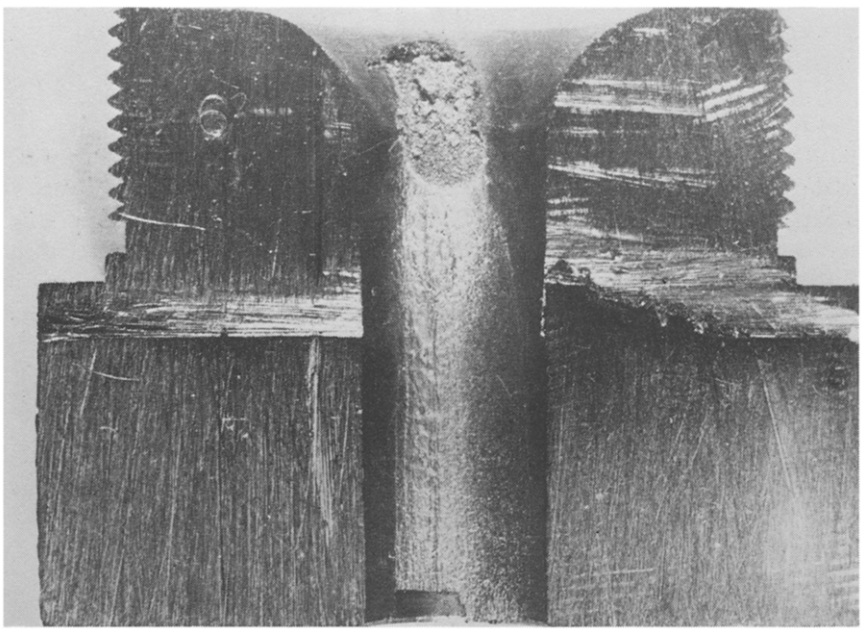


Fig. 23. Cerrobend nozzle melting from a choke fissure.

The tests indicated that in the absence of a fissure, pressure tapping hole or similar irregularity, initial damage takes place at the throat section with the formation of small pin-holes in the wall surface. Melting is then cumulative from these holes due to the eddies and turbulence created. Other tests indicated that local melting must first occur before material is removed. Apparently the shear forces created at the wall by the air stream are not sufficient to remove material weakened by elevated temperature before the melting point is reached.

In a further series of tests the effect of the initial surface temperature of the specimen on the erosion damage created by a particle-laden gas stream was investigated. Cerrobend alloy plates were mounted normal to the incident stream and, before particle attack, were preheated by the hot particle-free stream to various steady state temperatures below that of the melting point of the alloy. The results are given in Fig. 24. It will be seen that the erosion damage rises linearly with initial plate temperature.

The determination of particle efflux velocity

A photographic technique was used to obtain the velocity of the particles as they leave the nozzle. A short duration electronic flash unit is synchronized with a

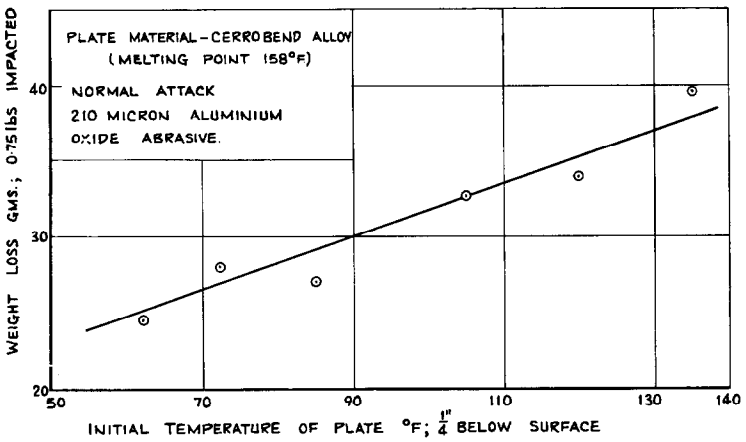


Fig. 24. Variation of erosion with initial plate temperature.

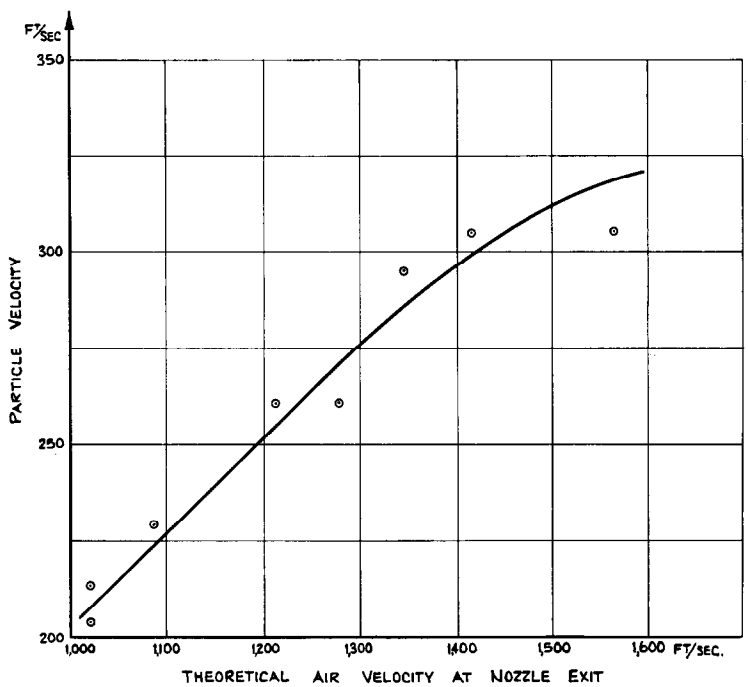


Fig. 25. Variation of particle velocity with theoretical air velocity for 500 μ aluminium oxide particles.

plate camera and supplies energy to two lamps with a predetermined time delay between the flashes at each lamp. Two separate exposures of a particle or group of particles are thus obtained on the same negative at a known time interval. Hence, by measuring the distance between the two positions of any specific particle or group, the velocity may be calculated. Particle velocities were obtained for different sizes and types of particles. The results obtained using 500 μ aluminium oxide particles for various air efflux velocites from a nozzle are shown in Fig. 25.

The variation in particle efflux velocity with the ratio of air mass flow rate to particle mass flow rate was investigated. The nozzle box pressure was kept constant while varying the flow rate of particles by using different sizes of control orifice plates. Using the photographic technique, the mean efflux velocity was obtained for each mass flow rate of particles and the results are shown in Fig. 26. It will be noted that the particle velocity remains constant until the largest control orifice plate is used when the particle velocity decreases.

As a check on this finding, further tests were made to determine the effect on erosion of flat plates of the mass flow rate of abrasive. The plates were each mounted at 40° to the incident stream and subjected to the abrasive action of the same weight

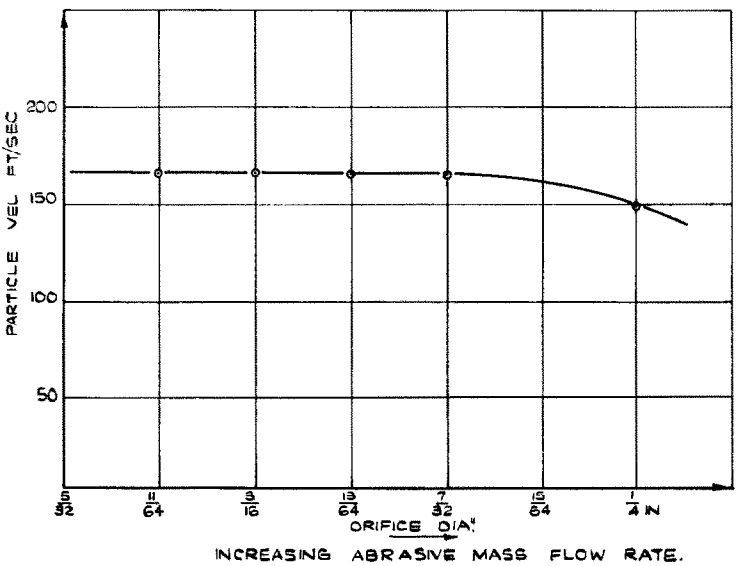


Fig. 26. Variation of particle velocity for 710 μ aluminium oxide particles with orifice diameter.

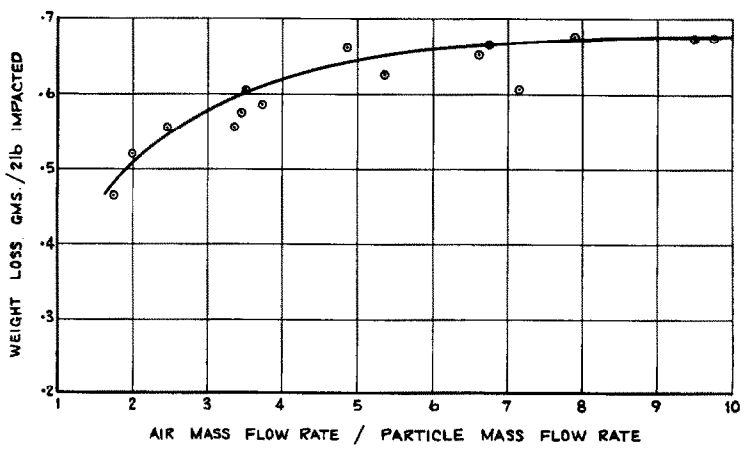


Fig. 27. Variation of erosion for aluminium plates with the ratio of air mass flow rate to particle mass flow rate for 210 μ aluminium oxide particles.

of charge, the mass flow rate of particles being varied from plate to plate. The results are shown in Fig. 27 where the loss of weight due to erosion is plotted against the ratio of air mass flow rate to particle mass flow rate. It will be seen that above a flow rate ratio of 6.5 the weight loss remains constant while below this critical value the erosion damage decreases. This confirms the previous result and it may be presumed that below the critical value the reduction in weight loss is due to a decrease in particle efflux velocity.

The assessment of the resistance offered to a particle attack by the "anti-erosion" coatings in current use in rocket motors

A number of mild steel plates were coated with thin layers (0.010 in.–0.030 in. thick) of alumina, Norton Rokide A, Norton Rokide Z (zirconia), Norton Rokide A4 and tungsten carbide. The plates were subjected to the action of quantities of 210 μ aluminium oxide particles at a nozzle box pressure of 25 p.s.i.g. It was found that one pound of abrasive charge was sufficient to remove the alumina and Rokide coatings and reveal the mild steel. Maximum damage occurred under a normal attack, the resistance increasing with decreasing angle of attack. For the tungsten carbide coating no erosion was detected.

CONCLUSIONS

The results of the investigation may be summarised under the following headings.

Pattern of damage

(a) At any cross-section in a nozzle even erosion results from an even distribution of solid particles in the gas stream. Where there is a concentration of particles at one angular position grooving may occur.

(b) Where the distribution of particles at entry to the nozzle is uneven the distribution will become progressively more even with distance through the nozzle.

Extent of damage

(a) For a given nozzle material, and abrasive particles of a given size and velocity, the extent of erosion damage depends on the angle of attack of the particles on the nozzle wall. Maximum damage is effected at a small angle of attack for a ductile material and under normal impact for a hard brittle material.

(b) Damage increases with increasing particle velocity and with decreasing particle size.

Factors affecting both the pattern and extent of damage

(a) In an actual rocket motor the speed, distribution and direction of the particles leaving the combustion chamber are of importance in determining the pattern and extent of damage.

(b) For a long blast pipe, where the ratio of blast tube area to nozzle throat area is small, the particle distribution leaving the tube will be even and the velocity will be high. Hence, in this case a definite even erosion of the upper part of the choke would be expected.

(c) Gouging of the nozzle, due to uneven flow distribution of the particles, has a cumulative adverse effect. Similar effects would result from a badly machined nozzle.

Mechanism of erosion

The fact that maximum erosion damage occurs for a ductile material at small angles of attack indicates that shear forces play some part in surface failure under bombardment from particles. Since, however, failure does not occur when the particle stream is moving parallel to the specimen, the criterion of failure must be some function of both shear forces and normal impact forces.

The experiments showing the apparent movement of the machined grooves within the nozzle passage indicate that, by using flat plates, one may effectively simulate the range of conditions to which the nozzle wall may be subjected. The angle for maximum damage appears to be independent of material properties, size or velocity of the abrasive particle. For brittle materials failure appears to be caused by cracking followed by crazing and subsequent spalling of material. For a ductile material plastic flow occurs and material is removed by the cutting action of the particle.

The outstanding feature of the plate surface temperature measurements is the high temperatures attained due to particle impact relative to the temperatures reached in a particle free stream. While a large temperature rise is observed, it is felt that the actual local surface temperature, as the particle makes contact, may be considerably higher than the indicated value.

Elevated temperature experiments

The tests, in which nozzles of low melting point material were subjected to the action of a hot, particle-free air stream, show that initial wall damage may be affected by the following factors.

(a) Irrègularities in the wall surface cause turbulence and so become foci of initial melting. Melting may then occur at gas stream temperatures lower than that at which melting would normally be initiated in the absence of irregularities.

(b) Where there are no irregularities the formation of pin-holes is the first sign of wall damage and melting is cumulative from these holes. The area first affected is just downstream of the throat section.

(c) Where the hot air stream contains particles, erosion damage increases as the wall surface temperature increases, the weight loss being a linear function of the initial surface temperature of the wall prior to the particle attack. Here, presumably, the elevated temperature reduces the mechanical strength of the material. The further increase in temperature during the actual attack will involve a further reduction in the strength of the material.

Determination of the particle efflux velocity

The experiments have shown that small particles have a much greater erosion potential than large particles. A hard steel nozzle, which had previously withstood the erosive action of considerable quantities of large sized iron and aluminium oxide particles, was easily damaged by the action of 290 μ and 210 μ aluminium oxide particles. Whether or not this is due entirely to the greater acceleration suffered by the smaller particles has not yet been established. The photographic technique for particle velocity determination has been applied, in the main, to particle sizes above 500 μ. Here the

particle velocity is shown to be much lower than the theoretical air velocity at the nozzle outlet.

There is ample evidence that the concentration of particles in the air stream is an important criterion in assessing the final particle velocity on which in turn the degree of erosion depends. It is shown that where the ratio of air mass flow rate to particle mass flow rate is greater than 6.5 the value of this ratio has no effect on the final efflux velocity. As the ratio is reduced below this critical figure, however, there is a continuous decrease in particle velocity and in the erosion damage sustained.

Tests on proprietary "anti-erosion" coatings in current use in rocket motor tail nozzles

The thin layers of alumina and the Rokides A, Z and A4 with which nozzle walls are covered, exhibit a very low resistance to even a moderate particle attack.

These investigations were of an exploratory nature and were followed by a fairly comprehensive analysis of the factors affecting the damage sustained in rocket nozzles by the action of gas-borne particles. The erosion at a given cross-section in a nozzle depends on the mass of particles striking the wall at that section, on the particle velocity, on the angle of attack and on the properties of the wall material.

A theoretical and experimental analysis of the erosion characteristics of different types of material has been given by the authors[1] in which reference is made to important work on this subject by FINNIE[2,3] and by BITTER[4]. The factors on which the velocity of a particle depends and methods for computing particle velocity in various gas streams are given in another study[5]. Determination of the angle of attack of a particle is described elsewhere[6].

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